

Quasi-static model of the chinese paper yo-yo

Brandon Wong, Ben Foster

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1 Geometric model

The system can be modelled as a constant-width rectangular ribbon of negligible thickness wrapped around a uniform-radius cylinder at a uniform pitch. One end of the ribbon is fixed in place to prevent rigid body motion, while the other (“active” end) is displaced in the z direction (the axis of the cylinder). Refer to Figure 1 for a visual representation of the system. The system has the following geometric constants:

- L : the length of the unrolled rectangle.
- W : the width of the unrolled rectangle. Assume $L \gg W$.
- θ_L : the total wind angle of the helix, which is fixed based on the boundary conditions (the two ends of the ribbon cannot rotate about the z -axis relative to one another, only translate along it, so the total number of turns is a fixed topological property of how the ribbon was wound and does not change as it is pulled). $\theta_L = 2\pi n$ where n is the number of winds around the z axis (not necessarily integer).

The state of the system is parametrized by the parameter x , which describes how far the active end has displaced in the z direction relative to the base end.

1.1 Coordinate systems

Positions on the helix can be described with local surface u, v coordinates, which describe the unrolled rectangle. The u direction runs along the length of the unrolled rectangle, and the v direction runs across its width. Alternatively, positions can be described in 3d space using cylindrical r, θ, z coordinates.

To simplify the geometry, we assume that the helix has a constant radius r and a constant pitch angle ϕ (the angle the local $\hat{u}$ makes with the r, θ plane). Equivalently, the local orthonormal frame $(\hat{u}, \hat{v})$ tangent to the ribbon surface is related to the cylinder’s own natural frame $(\hat{\theta}, \hat{z})$ by a rotation through the angle ϕ :

$$\hat{u} = \cos \phi \hat{\theta} + \sin \phi \hat{z}, \quad \hat{v} = -\sin \phi \hat{\theta} + \cos \phi \hat{z}. \quad (1)$$

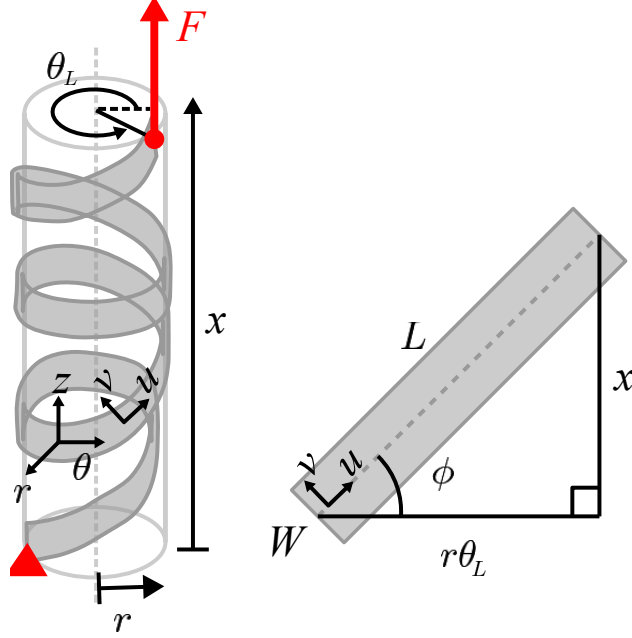


Figure 1: in progress

We now use Eq. (1) to relate the geometric constants L, θ_L to the pitch angle ϕ and radius r at a given extension x . From Eq. (1), $\hat{u}$ has z -component $\sin \phi$, so moving a distance du along the ribbon's own length produces an axial displacement $dz = \sin \phi du$. Since ϕ is constant along the whole ribbon at a given (frozen) extension x , integrating this from the base end ($u = 0$) to the active end ($u = L$) gives a total axial displacement of $L \sin \phi$. By definition, this total axial displacement between the two ends of the ribbon is exactly x , so

$$x = L \sin \phi \quad \implies \quad \phi = \sin^{-1} \left(\frac{x}{L} \right). \quad (2)$$

Likewise, $\hat{u}$ has θ -component $\cos \phi$, so moving a distance du along the ribbon produces a circumferential arc length $r d\theta = \cos \phi du$. Integrating over the whole ribbon length L , the ribbon sweeps out a total circumferential arc length $L \cos \phi$. This total arc length must also equal $r \theta_L$ (arc length equals radius times swept angle), since by construction the ribbon winds through the fixed total angle θ_L regardless of x . Equating the two expressions for this arc length and using $\cos \phi = \sqrt{1 - \sin^2 \phi}$ together with Eq. (2) gives

$$r \theta_L = L \cos \phi \quad \implies \quad r = \frac{L}{\theta_L} \cos \phi = \frac{L}{\theta_L} \sqrt{1 - \frac{x^2}{L^2}} = \frac{\sqrt{L^2 - x^2}}{\theta_L}. \quad (3)$$

1.2 Position vector

An area element $du \times dv$ at point (u, v) has a position vector given in cylindrical coordinates. We build this up from Eq. (1): since $\hat{u} = \cos \phi \hat{\theta} + \sin \phi \hat{z}$, moving along the ribbon by du produces $r d\theta = \cos \phi du$ and $dz = \sin \phi du$; since $\hat{v} = -\sin \phi \hat{\theta} + \cos \phi \hat{z}$, moving across the width by dv produces $r d\theta = -\sin \phi dv$ and $dz = \cos \phi dv$. Integrating these (at fixed, frozen x , so r and ϕ are

constants) from the reference point $(u, v) = (0, 0)$ gives

$$\vec{p}(u, v, x) = \begin{bmatrix} r \\ \theta \\ z \end{bmatrix} = \begin{bmatrix} r \\ \frac{u \cos \phi - v \sin \phi}{r} \\ u \sin \phi + v \cos \phi \end{bmatrix}. \quad (4)$$

1.3 Local curvature

A cylinder of radius r has two principal curvatures: $\kappa_\theta = 1/r$ in the circumferential ($\hat{\theta}$) direction in which it is rolled, and $\kappa_z = 0$ in the axial ($\hat{z}$) direction in which it is completely flat. Equivalently, the standard cylindrical basis vectors rotate with θ at the rates

$$\frac{\partial \hat{r}}{\partial \theta} = \hat{\theta}, \quad \frac{\partial \hat{\theta}}{\partial \theta} = -\hat{r}, \quad \frac{\partial \hat{z}}{\partial \theta} = 0, \quad (5)$$

(the first two follow from writing $\hat{r} = \cos \theta \hat{x} + \sin \theta \hat{y}$ and $\hat{\theta} = -\sin \theta \hat{x} + \cos \theta \hat{y}$ in terms of the fixed Cartesian basis and differentiating; the third holds because $\hat{z}$ does not depend on θ at all — this is exactly the statement $\kappa_z = 0$).

To differentiate the ribbon's own frame $(\hat{u}, \hat{v}, \hat{r})$ with respect to (u, v) at a fixed x (and therefore a fixed r, ϕ), we take the partial derivatives of the θ component of Eq. (4):

$$\frac{\partial \theta}{\partial u} = \frac{\cos \phi}{r}, \quad \frac{\partial \theta}{\partial v} = -\frac{\sin \phi}{r}. \quad (6)$$

Combining Eq. (5) and Eq. (6) via the chain rule (e.g. $\frac{\partial \hat{r}}{\partial u} = \frac{\partial \hat{r}}{\partial \theta} \frac{\partial \theta}{\partial u}$) gives

$$\frac{\partial \hat{r}}{\partial u} = \frac{\cos \phi}{r} \hat{\theta}, \quad \frac{\partial \hat{r}}{\partial v} = -\frac{\sin \phi}{r} \hat{\theta}, \quad \frac{\partial \hat{\theta}}{\partial u} = -\frac{\cos \phi}{r} \hat{r}, \quad \frac{\partial \hat{\theta}}{\partial v} = \frac{\sin \phi}{r} \hat{r}, \quad (7)$$

with $\hat{z}$ constant with respect to u, v . Now we take the partial derivatives of $\hat{u}, \hat{v}$ themselves, using Eq. (1) with ϕ held fixed:

$$\begin{aligned} \frac{\partial \hat{u}}{\partial u} &= \cos \phi \frac{\partial \hat{\theta}}{\partial u} = -\frac{\cos^2 \phi}{r} \hat{r}, & \frac{\partial \hat{u}}{\partial v} &= \cos \phi \frac{\partial \hat{\theta}}{\partial v} = \frac{\sin \phi \cos \phi}{r} \hat{r}, \\ \frac{\partial \hat{v}}{\partial u} &= -\sin \phi \frac{\partial \hat{\theta}}{\partial u} = \frac{\sin \phi \cos \phi}{r} \hat{r}, & \frac{\partial \hat{v}}{\partial v} &= -\sin \phi \frac{\partial \hat{\theta}}{\partial v} = -\frac{\sin^2 \phi}{r} \hat{r}. \end{aligned} \quad (8)$$

The same three coefficients, $\cos^2 \phi/r$, $\sin^2 \phi/r$, and $\sin \phi \cos \phi/r$ (the last one twice), keep reappearing. We name them for convenience:

$$\kappa_{uu} := \frac{\cos^2 \phi}{r}, \quad \kappa_{vv} := \frac{\sin^2 \phi}{r}, \quad \kappa_{uv} := \frac{\sin \phi \cos \phi}{r}, \quad (9)$$

Finally, we differentiate $\hat{r}$ itself. We already have $\frac{\partial \hat{r}}{\partial u}, \frac{\partial \hat{r}}{\partial v}$ in terms of $\hat{\theta}$ (Eq. 7). To express them in terms of $\hat{u}, \hat{v}$ instead, we invert Eq. (1) to obtain $\hat{\theta} = \cos \phi \hat{u} - \sin \phi \hat{v}$. Substituting this

expression for $\hat{\theta}$ into Eq. (7) gives

$$\frac{\partial \hat{r}}{\partial u} = \frac{\cos \phi}{r} (\cos \phi \hat{u} - \sin \phi \hat{v}) = \kappa_{uu} \hat{u} - \kappa_{uv} \hat{v} \quad (10)$$

$$\frac{\partial \hat{r}}{\partial v} = -\frac{\sin \phi}{r} (\cos \phi \hat{u} - \sin \phi \hat{v}) = -\kappa_{uv} \hat{u} + \kappa_{vv} \hat{v} \quad (11)$$

Collecting everything derived in this section,

$$\begin{aligned} \frac{\partial \hat{r}}{\partial u} &= \kappa_{uu} \hat{u} - \kappa_{uv} \hat{v}, & \frac{\partial \hat{r}}{\partial v} &= -\kappa_{uv} \hat{u} + \kappa_{vv} \hat{v}, \\ \frac{\partial \hat{u}}{\partial u} &= -\kappa_{uu} \hat{r}, & \frac{\partial \hat{u}}{\partial v} &= \kappa_{uv} \hat{r}, \\ \frac{\partial \hat{v}}{\partial u} &= \kappa_{uv} \hat{r}, & \frac{\partial \hat{v}}{\partial v} &= -\kappa_{vv} \hat{r}. \end{aligned} \quad (12)$$

2 Stress balance

We will model the paper yoyo as a thin shell of small but finite thickness $t \ll r$. Because the ribbon's thickness ultimately carries the boundary tractions responsible for friction (Section 3), we work throughout with its full three-dimensional stress state. We use $\zeta \in [-t/2, t/2]$ for the coordinate measuring position through the thickness, i.e., displacement from the mid-surface along $\hat{r}$. At a point (u, v, ζ) the material carries the full Cauchy stress tensor:

$$\sigma = \begin{bmatrix} \sigma_{uu} & \sigma_{uv} & \sigma_{ur} \\ \sigma_{uv} & \sigma_{vv} & \sigma_{vr} \\ \sigma_{ur} & \sigma_{vr} & \sigma_{rr} \end{bmatrix} \quad (13)$$

There are six independent components: the in-plane stresses $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}$; the transverse shears σ_{ur}, σ_{vr} by which the ribbon can transmit a load sideways from one part of its width to another, and the normal pressure σ_{rr} caused by layer-layer contact at the ribbon's surface.

We perform a force balance on a differential volume element $du \times dv \times d\zeta$. Each direction has a corresponding traction vector:

$$\vec{t}_u = \sigma_{uu} \hat{u} + \sigma_{uv} \hat{v} + \sigma_{ur} \hat{r} \quad \vec{t}_v = \sigma_{uv} \hat{u} + \sigma_{vv} \hat{v} + \sigma_{vr} \hat{r} \quad \vec{t}_\zeta = \sigma_{ur} \hat{u} + \sigma_{vr} \hat{v} + \sigma_{rr} \hat{r} \quad (14)$$

Because $t \ll r$, the element can be approximated as a rectangular prism, so the net force on the element is

$$\left(\frac{\partial \vec{t}_u}{\partial u} + \frac{\partial \vec{t}_v}{\partial v} + \frac{\partial \vec{t}_\zeta}{\partial \zeta} \right) du dv d\zeta = 0 \quad (15)$$

However, because $\hat{u}, \hat{v}, \hat{r}$ themselves depend on u, v (Eq. 12), differentiating $\vec{t}_u, \vec{t}_v, \vec{t}_\zeta$ produces extra terms beyond the plain derivatives of the stress components. For example, expanding $\frac{\partial}{\partial u}(\sigma_{uu} \hat{u})$ introduces a curvature-weighted term in the $\hat{r}$ direction:

$$\frac{\partial}{\partial u}(\sigma_{uu} \hat{u}) = \frac{\partial \sigma_{uu}}{\partial u} \hat{u} + \sigma_{uu} \frac{\partial \hat{u}}{\partial u} = \frac{\partial \sigma_{uu}}{\partial u} \hat{u} - \kappa_{uu} \sigma_{uu} \hat{r} \quad (16)$$

The additional curvature-weighted terms reflect the fact that the element is not perfectly planar, so in-plane stresses are slightly rotated to have a small out-of-plane component, and vice versa. Expanding all terms and sorting by component gives three scalar force-balance equations:

$$\hat{u} : \boxed{\frac{\partial \sigma_{uu}}{\partial u} + \frac{\partial \sigma_{uv}}{\partial v} + \frac{\partial \sigma_{ur}}{\partial \zeta} + \kappa_{uu}\sigma_{ur} - \kappa_{uv}\sigma_{vr} = 0} \quad (17)$$

$$\hat{v} : \boxed{\frac{\partial \sigma_{uv}}{\partial u} + \frac{\partial \sigma_{vv}}{\partial v} + \frac{\partial \sigma_{vr}}{\partial \zeta} - \kappa_{uv}\sigma_{ur} + \kappa_{vv}\sigma_{vr} = 0} \quad (18)$$

$$\hat{r} : \boxed{\frac{\partial \sigma_{ur}}{\partial u} + \frac{\partial \sigma_{vr}}{\partial v} + \frac{\partial \sigma_{rr}}{\partial \zeta} - \kappa_{uu}\sigma_{uu} + 2\kappa_{uv}\sigma_{uv} - \kappa_{vv}\sigma_{vv} = 0} \quad (19)$$

Equilibrium alone only gives 3 equations for 6 unknown stress components. However, because t is small, we will make a first-order approximation that the stress field varies linearly through t . This means that for any stress component, if we know its boundary value at both $-\frac{t}{2}, \frac{t}{2}$ for all (u, v) , that component is essentially solved. In the following section, we will obtain the surface boundary conditions for σ_{ur}, σ_{vr} , and σ_{rr} based on layer-layer contact.

3 Friction

3.1 Overlapping neighbors

Because the ribbon is wound around itself, a given area element at (u, v) could physically overlap with another area elements one full turn away along the ribbon. Specifically, an area element at point (u, v) overlaps with the neighboring interior layer at point (u^-, v^-) if their cylindrical coordinates have equal r and z , but differ by one full turn in θ . We can express this by taking the difference in $\vec{p}$:

$$\vec{p}(u, v, x) - \vec{p}(u^-, v^-, x) = \begin{bmatrix} 0 \\ 2\pi \\ 0 \end{bmatrix} \quad (20)$$

Plugging in Eq (4) gives:

$$u^- = u - 2\pi r \cos \phi, \quad (21)$$

$$v^- = v + 2\pi r \sin \phi. \quad (22)$$

By the identical argument with $\theta(u^+, v^+) = \theta(u, v) + 2\pi$ (i.e., the neighboring *outer* layer, one full turn further out), the element has an outer overlapping neighbor at

$$u^+ = u + 2\pi r \cos \phi, \quad (23)$$

$$v^+ = v - 2\pi r \sin \phi. \quad (24)$$

Some points (u, v) on the surface overlap with an inner or outer layer, while others don't. We

neglect the edge effects where a neighbor is absent because u^-, u^+ is outside the $[0, L]$ bounds. As a result, an inner neighbor is present exactly whenever v^- lies within $[-W/2, W/2]$, and an outer neighbor is present exactly whenever v^+ lies in $[-W/2, W/2]$. That is,

$$\text{Inner neighbor is present if } v \leq \frac{W}{2} - 2\pi r \sin \phi \quad (25)$$

$$\text{Outer neighbor is present if } v \geq -\frac{W}{2} + 2\pi r \sin \phi \quad (26)$$

3.2 Friction direction

The position vector $\vec{p}(u, v, x)$ of a material point (u, v) is given by Eq. (4). Substituting Eq. (2) and Eq. (3) to expose the dependencies on x gives:

$$\vec{p}(u, v, x) = \begin{bmatrix} \frac{\sqrt{L^2 - x^2}}{\theta_L} \\ \frac{u\theta_L}{L} - \frac{v x \theta_L}{L\sqrt{L^2 - x^2}} \\ \frac{ux}{L} + \frac{v\sqrt{L^2 - x^2}}{L} \end{bmatrix} \quad (27)$$

The velocity of a given material point (u, v) as the active end of the ribbon is displaced (i.e., as x changes) is then obtained by taking the partial derivative of each component of Eq. (4) with respect to x :

$$\frac{\partial}{\partial x} \vec{p}(u, v, x) = \begin{bmatrix} \frac{\partial r}{\partial x} \\ \frac{\partial \theta}{\partial x} \\ \frac{\partial z}{\partial x} \end{bmatrix} = \begin{bmatrix} -\frac{x}{\theta_L \sqrt{L^2 - x^2}} \\ -\frac{L \theta_L v}{(L^2 - x^2)^{3/2}} \\ \frac{u}{L} - \frac{vx}{L\sqrt{L^2 - x^2}} \end{bmatrix}. \quad (28)$$

We can then plug u^-, v^- into Eq. (28) to compute the velocity of the element relative to its inner overlapping neighbor, which gives the relative slip direction in the r, θ, z frame:

$$\frac{\partial \vec{p}(u, v, x)}{\partial x} - \frac{\partial \vec{p}(u^-, v^-, x)}{\partial x} = \begin{bmatrix} 0 \\ \frac{2\pi x}{L^2 - x^2} \\ \frac{2\pi}{\theta_L} \end{bmatrix}. \quad (29)$$

As expected, there is no relative velocity in the radial direction (i.e., the layers are not lifting away from each other). We can now compute the relative slip direction in the local frame by projecting this slip velocity from the θ, z basis onto the u, v basis using the relation described by Eq. (1). Since $\frac{\partial \theta}{\partial x}$ above is an *angular* rate, we first convert it to a linear (arc-length) rate by multiplying by r , giving $v_\theta = r \frac{\partial \theta}{\partial x}$; then projecting as just described:

$$\vec{v}_{slip}(u, v, x) = \begin{bmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} r \cdot \frac{2\pi x}{L^2 - x^2} \\ \frac{2\pi}{\theta_L} \end{bmatrix} \quad (30)$$

We simplify this expression by substituting x and r in terms of ϕ (Eq. (2), (3)) to expose the double-angle combination, then applying double angle trig identities to factor out

common terms. The result is a scaled unit vector:

$$\vec{v}_{slip}(u, v, x) = \frac{2\pi}{\theta_L \cos \phi} \begin{bmatrix} \sin(2\phi) \\ \cos(2\phi) \end{bmatrix} \quad (31)$$

The prefactor is strictly positive for all $x \in (0, L)$. Therefore, the relative slip velocity points in the direction of the unit vector $-(\sin 2\phi, \cos 2\phi)$ in the local u, v frame, so the Coulomb friction force opposing this motion must point in the opposite direction:

$$\hat{f} = - \begin{bmatrix} \sin(2\phi) \\ \cos(2\phi) \end{bmatrix} \quad (32)$$

3.3 Friction magnitude

Let us take a moment to introduce some shorthand notation:

$$\sigma_{rr}^{\pm} := \sigma_{rr} \left(\zeta = \pm \frac{t}{2} \right) \quad (33)$$

$$\Delta\sigma_{rr} := \sigma_{rr}^+ - \sigma_{rr}^- \quad (34)$$

$$S_{rr} := \sigma_{rr}^+ + \sigma_{rr}^- \quad (35)$$

Recall that Eq. 32 is the friction exerted on the element by the inner layer, which resists the element from sliding forward with x . The components of this friction force are equivalent to the shear stresses acting on the inner ($-\hat{r}$) face. Additionally, the magnitude of the Coulomb friction force is equivalent to the normal pressure $-\sigma_{rr}$ times a friction coefficient μ . By saying this, we assume $\sigma_{rr} \leq 0$ (the element is under compression), deferring a rigorous justification to Section 4.3.

$$\sigma_{ur} \left(\zeta = -\frac{t}{2} \right) = -\mu\sigma_{rr}^- \sin(2\phi) \quad (36)$$

$$\sigma_{vr} \left(\zeta = -\frac{t}{2} \right) = -\mu\sigma_{rr}^- \cos(2\phi) \quad (37)$$

By the same argument, the friction exerted by the outer layer is the exact opposite, dragging the element forward. We then obtain a similar boundary relationship for the outer ($+\hat{r}$) face. Note that the sign is positive due to the sign convention for shear stresses on the inner face of an element.

$$\sigma_{ur} \left(\zeta = \frac{t}{2} \right) = \mu\sigma_{rr}^+ \sin(2\phi) \quad (38)$$

$$\sigma_{vr} \left(\zeta = \frac{t}{2} \right) = \mu\sigma_{rr}^+ \cos(2\phi) \quad (39)$$

Let thickness-averaged stress components be denoted by a bar, i.e., $\bar{\sigma}_{ij} := \frac{1}{t} \int_{-\frac{t}{2}}^{\frac{t}{2}} \sigma_{ij} d\zeta$. For simplicity, we assume the thickness is small enough that the variation of all stress components through the thickness can be approximated as linear. We can now use Eqs. 36–39 to get the

thickness-averaged values $\bar{\sigma}_{ur}$ and $\bar{\sigma}_{vr}$ in terms of the two boundary values of σ_{rr} :

$$\boxed{\bar{\sigma}_{ur} = -\frac{\mu}{2} S_{rr} \sin(2\phi)} \quad (40)$$

$$\boxed{\bar{\sigma}_{vr} = -\frac{\mu}{2} S_{rr} \cos(2\phi)} \quad (41)$$

3.4 Layer-layer contact relationship

By Newton's third law, two surfaces in contact should experience equal and opposite traction forces. Because the surface normals are opposite, this means they should have equal stresses. It suffices to equate the normal pressures σ_{rr} because the transverse shear stresses are related above. Therefore, we get the following recursive relationships to define the boundary stresses, using piecewise gates based on layer contact boundaries:

$$\text{Outer surface: } \sigma_{rr}^+(u, v) = \begin{cases} \sigma_{rr}^-(u^+, v^+) & v \geq -\frac{W}{2} + 2\pi r \sin \phi \\ 0 & \text{otherwise} \end{cases} \quad (42)$$

$$\text{Inner surface: } \sigma_{rr}^-(u, v) = \begin{cases} \sigma_{rr}^+(u^-, v^-) & v \leq \frac{W}{2} - 2\pi r \sin \phi \\ 0 & \text{otherwise} \end{cases} \quad (43)$$

$$(44)$$

The above relationships recursively define the boundary stresses of the entire ribbon, using the boundary condition at the outermost layer $\sigma_{rr}(u = L) = 0$ as a base case.

4 PDE formulation and closed-form solution

We now assemble the pieces above into a closed system for the stress field. The three equilibrium equations, Eqs. (17)–(19), are three PDEs for six unknown stress components at every point (u, v, ζ) . Section 3 supplies what is needed to close this at the two surfaces $\zeta = \pm \frac{t}{2}$: there, σ_{ur}, σ_{vr} are fixed directly in terms of σ_{rr} by Coulomb friction (Eqs. 36–39), and σ_{rr} 's own two boundary values are tied recursively, layer to layer, by Newton's third law (Eqs. 42–43). Combined with the approximation that the stress field varies linearly through the thickness, this eliminates the unknown $\sigma_{rr}, \sigma_{ur}, \sigma_{vr}$ components and reduces the remaining unknowns to a function of (u, v) alone.

4.1 In-plane thickness integration

We now integrate the three equilibrium equations through the thickness. Beginning with the $\hat{u}$ direction Eq (17),

$$\frac{1}{t} \int_{-\frac{t}{2}}^{\frac{t}{2}} \left(\frac{\partial \sigma_{uu}}{\partial u} + \frac{\partial \sigma_{uv}}{\partial v} + \frac{\partial \sigma_{ur}}{\partial \zeta} + \kappa_{uu} \sigma_{ur} - \kappa_{uv} \sigma_{vr} \right) d\zeta = 0 \quad (45)$$

$$\frac{\partial \bar{\sigma}_{uu}}{\partial u} + \frac{\partial \bar{\sigma}_{uv}}{\partial v} + \frac{1}{t} \sigma_{ur} \Big|_{-\frac{t}{2}}^{\frac{t}{2}} + \kappa_{uu} \bar{\sigma}_{ur} - \kappa_{uv} \bar{\sigma}_{vr} = 0 \quad (46)$$

Then substituting Eqs. (40)–(41) for the thickness-averaged transverse shear stresses, and expanding the trig identities in the curvature terms:

$$\frac{\partial \bar{\sigma}_{uu}}{\partial u} + \frac{\partial \bar{\sigma}_{uv}}{\partial v} - \frac{\mu \sin(2\phi)}{t} (\sigma_{rr}^+ - \sigma_{rr}^-) + \frac{\mu}{2} S_{rr} (\kappa_{uv} \cos(2\phi) - \kappa_{uu} \sin(2\phi)) = 0 \quad (47)$$

$$\frac{\partial \bar{\sigma}_{uu}}{\partial u} + \frac{\partial \bar{\sigma}_{uv}}{\partial v} - \frac{\mu \sin(2\phi) \Delta \sigma_{rr}}{t} - \frac{\mu \sin(2\phi) S_{rr}}{4r} = 0 \quad (48)$$

Integrating the $\hat{v}$ direction Eq (18) gives a similar result:

$$\frac{\partial \bar{\sigma}_{uv}}{\partial u} + \frac{\partial \bar{\sigma}_{vv}}{\partial v} - \frac{\mu \cos(2\phi) \Delta \sigma_{rr}}{t} + \frac{\mu \cos(2\phi) S_{rr}}{4r} = 0 \quad (49)$$

4.2 Out-of-plane thickness integration

Next, we integrate the $\hat{r}$ direction Eq (19):

$$\frac{1}{t} \int_{-\frac{t}{2}}^{\frac{t}{2}} \left(\frac{\partial \sigma_{ur}}{\partial u} + \frac{\partial \sigma_{vr}}{\partial v} + \frac{\partial \sigma_{rr}}{\partial \zeta} - \kappa_{uu} \sigma_{uu} + 2\kappa_{uv} \sigma_{uv} - \kappa_{vv} \sigma_{vv} \right) d\zeta = 0 \quad (50)$$

$$\frac{\partial \bar{\sigma}_{ur}}{\partial u} + \frac{\partial \bar{\sigma}_{vr}}{\partial v} + \frac{1}{t} \Delta \sigma_{rr} - \kappa_{uu} \bar{\sigma}_{uu} + 2\kappa_{uv} \bar{\sigma}_{uv} - \kappa_{vv} \bar{\sigma}_{vv} = 0 \quad (51)$$

Substituting Eqs. (40)–(41) for $\bar{\sigma}_{ur}, \bar{\sigma}_{vr}$ gives,

$$\boxed{\Delta \sigma_{rr} = t [\kappa_{uu} \sigma_{uu} - 2\kappa_{uv} \sigma_{uv} + \kappa_{vv} \sigma_{vv}] + \frac{t\mu}{2} \left[\sin(2\phi) \frac{\partial S}{\partial u} + \cos(2\phi) \frac{\partial S}{\partial v} \right]}. \quad (52)$$

This is one equation in the two unknowns $S_{rr}, \Delta \sigma_{rr}$ (equivalently, $\sigma_{rr}^+, \sigma_{rr}^-$); by itself it is not yet closed. We now bring in the recursive contact relationship, Eqs. (42)–(43), the same way Section ?? did for σ_{rr}^+ alone: Taylor-expand $\sigma_{rr}^-(u^+, v^+)$ about (u, v) in the fixed shift $(2\pi r \cos \phi, -2\pi r \sin \phi)$,

$$\sigma_{rr}^+(u, v) = \sigma_{rr}^-(u^+, v^+) \approx \sigma_{rr}^-(u, v) + 2\pi r \cos \phi \frac{\partial \sigma_{rr}^-}{\partial u} - 2\pi r \sin \phi \frac{\partial \sigma_{rr}^-}{\partial v}, \quad (53)$$

and, since $\sigma_{rr}^+ - \sigma_{rr}^-$ is already the small quantity being solved for, replace $\partial\sigma_{rr}^-/\partial u, \partial\sigma_{rr}^-/\partial v$ by $\frac{1}{2}\partial S/\partial u, \frac{1}{2}\partial S/\partial v$ at leading order (dropping the $O(\sigma_{rr}^+ - \sigma_{rr}^-)$ correction to this replacement itself):

$$\boxed{\sigma_{rr}^+ - \sigma_{rr}^- \approx \pi r \cos \phi \frac{\partial S}{\partial u} - \pi r \sin \phi \frac{\partial S}{\partial v}.} \quad (54)$$

Equations (52) and (54) are two independent expressions for the same quantity, $\sigma_{rr}^+ - \sigma_{rr}^-$, both written in terms of $\partial S/\partial u, \partial S/\partial v$. Equating them,

$$\left(\pi r \cos \phi - \frac{t\mu}{2} \sin(2\phi) \right) \frac{\partial S}{\partial u} - \left(\pi r \sin \phi + \frac{t\mu}{2} \cos(2\phi) \right) \frac{\partial S}{\partial v} = t[\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} + \kappa_{vv}\sigma_{vv}], \quad (55)$$

the $t\mu$ corrections to the two coefficients are smaller than the leading $\pi r \cos \phi, \pi r \sin \phi$ terms by a factor of order t/r (comparing an explicit t against an explicit r), so at leading order they drop, leaving the closed transport equation

$$\boxed{\pi r \cos \phi \frac{\partial S}{\partial u} - \pi r \sin \phi \frac{\partial S}{\partial v} = t[\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} + \kappa_{vv}\sigma_{vv}]} \quad (56)$$

for S alone — exactly twice Eq. (??) written for σ_{rr}^+ , consistent with $S \approx 2\sigma_{rr}^+$ once $\sigma_{rr}^+ - \sigma_{rr}^-$ is small. Under the width-uniform ansatz of Section ?? (S independent of v), integrating Eq. (56) along its characteristic out to $u' = L$ gives

$$S(u) = S(L) - \frac{t}{\pi r \cos \phi} \int_u^L [\kappa_{uu}\sigma_{uu}(u') - 2\kappa_{uv}\sigma_{uv}(u') + \kappa_{vv}\sigma_{vv}(u')] du'. \quad (57)$$

The boundary value $S(L)$ is fixed using $\sigma_{rr}^+(L, v) = 0$ (Section 3.4) together with Eq. (52), which at $u = L$ (dropping its own $t\mu$ term, consistently) gives $\sigma_{rr}^+(L) - \sigma_{rr}^-(L) \approx t[\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} + \kappa_{vv}\sigma_{vv}](L)$, so $S(L) = \sigma_{rr}^+(L) + \sigma_{rr}^-(L) = -t[\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} + \kappa_{vv}\sigma_{vv}](L)$. Recovering $\sigma_{rr}^+ = \frac{1}{2}(S + (\sigma_{rr}^+ - \sigma_{rr}^-))$, with the difference again given by Eq. (52) at leading order, gives the corrected closed form

$$\boxed{\sigma_{rr}^+(u) = -\frac{t}{2\pi r \cos \phi} \int_u^L [\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} + \kappa_{vv}\sigma_{vv}](u') du' + \frac{t}{2} [\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} + \kappa_{vv}\sigma_{vv}] \Big|_u^L,} \quad (58)$$

where $[\dots]_u^L$ denotes the bracket evaluated at $u' = L$ minus its value at $u' = u$. The first term reproduces Eq. (??) exactly; the second is the new piece surfaced by keeping (rather than dropping) the bracket in Eq. (??). It carries the same explicit factor of t as the first term but no compensating $1/r$, so it is smaller than the first term by a factor of order r/L (the number of turns) — small in the many-turns regime already assumed elsewhere, but now a derived, quantified correction rather than an asserted one.

4.3 Sign and monotonicity of the contact pressure

5 Experimental validation

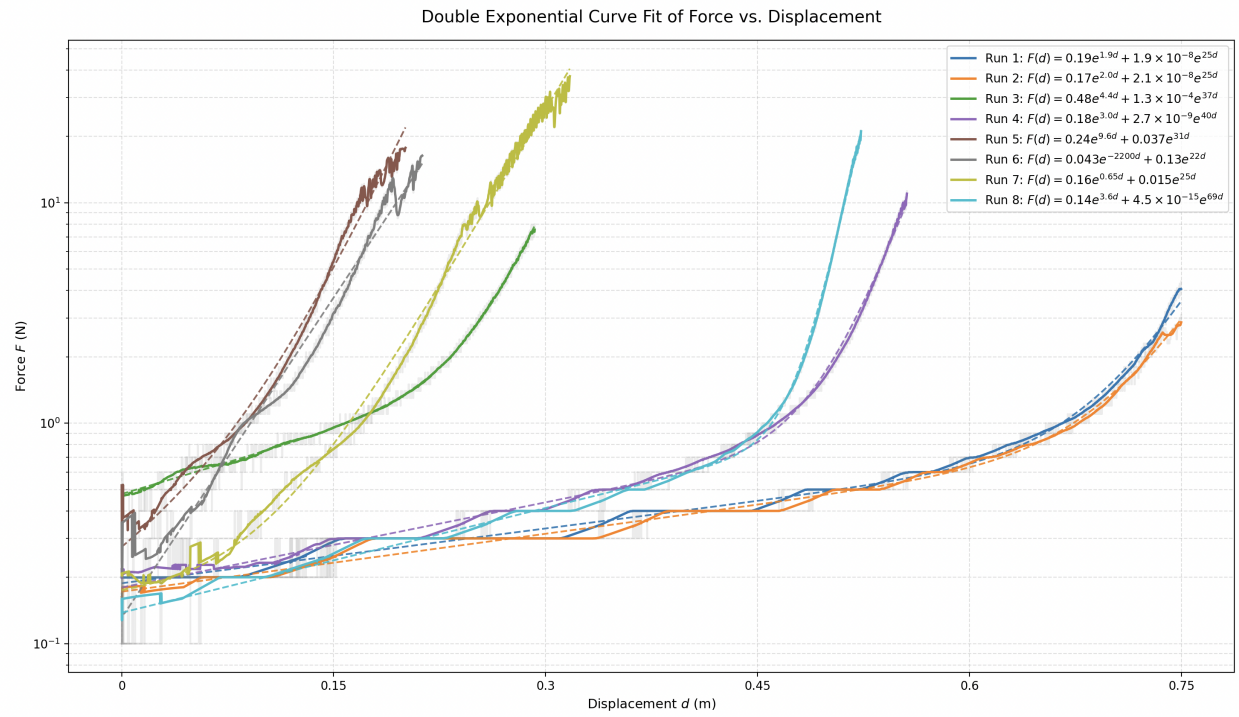


Figure 2: Instron force-displacement data, with raw quantized data (grey), smoothed data (solid colored), and curve fit (dashed colored).